Project Proposal: Space-Time Travel Planner

I. Project Overview

- **Project Name:** Space-Time Travel Planner
- **Proponent:** Lyan Ezekiel P. Tolentino (Section 8-Camia)
- **Subject / Domain:** Computer Science & Interplanetary Physics Simulation
- **Academic Institution:** Philippine Science High School – Eastern Visayas Campus

II. Problem Statement

Albert Einstein's Theory of General Relativity established that massive celestial bodies warp spacetime, dictating that gravitational fields and relative velocities directly influence physical metrics such as weight and temporal progression (Einstein, 1915; Thorne, 1994). When exploring interplanetary transit across different bodies in our solar system—such as Mars, Jupiter, and Venus—varying gravitational constants ($\text{Mars} = 0.38g$, $\text{Jupiter} = 2.34g$, $\text{Venus} = 0.90g$) drastically alter a traveler's effective weight, while extended orbital travel durations shift chronological age relative to Earth years.

Despite these fundamental principles of planetary physics, students and casual space enthusiasts often struggle to visualize or compute these cosmic differentials manually. Complex formulas involving transit windows, calendar increments, and planetary mass ratios can feel abstract and disconnected from practical application. This project addresses the gap by translating theoretical astrophysics into an accessible, interactive digital simulation tool that computes real-time interplanetary weight adjustments, travel durations, arrival dates, and age scaling dynamically based on user input.

III. Project Objectives

General Objective

To design, develop, and test an interactive software program that simulates interplanetary travel logistics, calculating planetary weight variations, transit arrival schedules, and adjusted ages based on celestial gravitational metrics.

Specific Objectives (SMART Framework)

- **Specific (S):** Program a terminal-based application using Python that accepts user weight and age inputs, provides choices across three distinct planetary destinations (Mars, Jupiter, and Venus), and applies specific gravitational multipliers and transit durations.
- **Measurable (M):** Accurately compute and display precise future arrival dates using Python's datetime module, alongside exact planetary weights (rounded to 2 decimal places) and adjusted ages (rounded to 1 decimal place).

- **Achievable (A):** Utilize fundamental programming constructs such as conditional selection structures (if-elif-else), arithmetic operations, and date-time arithmetic matching Grade 8 Computer Science coursework.
- **Relevant (R):** Bridge the intersection of Earth science and computer science by turning abstract celestial mechanics and gravitational physics into a tangible, user-friendly utility.
- **Time-bound (T):** Complete the code implementation, pseudocode, flowchart modeling, and proposal documentation within the prescribed academic quarter timeline.